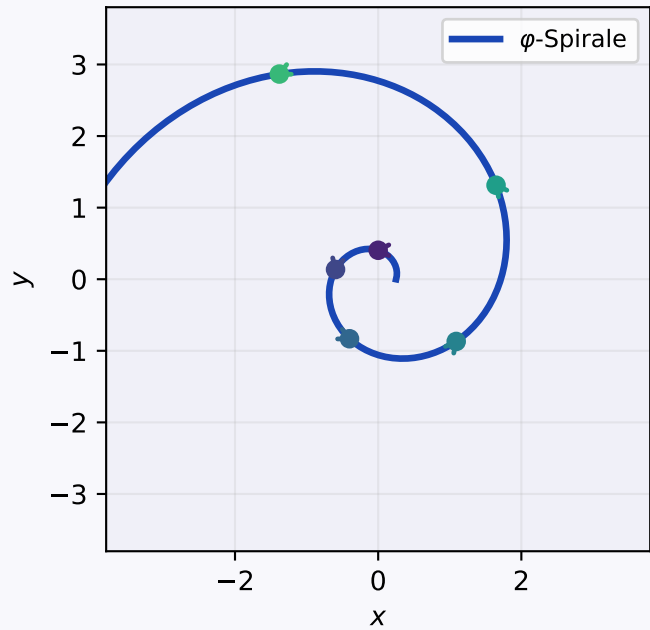
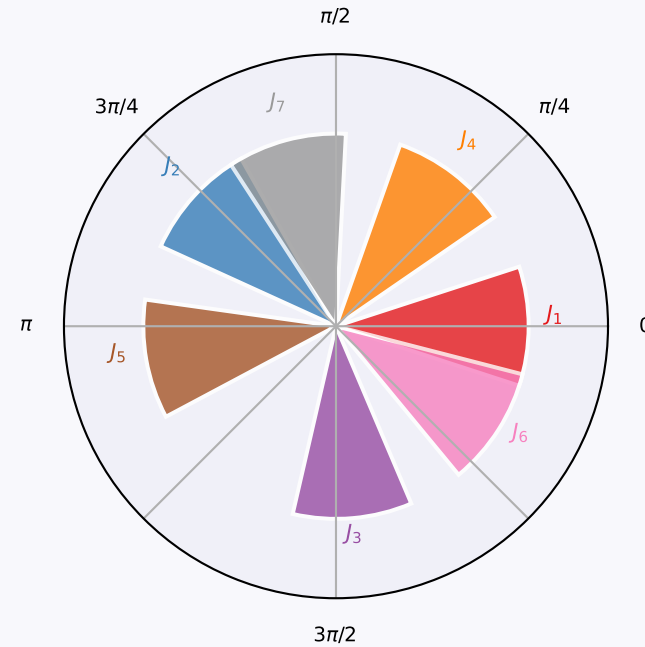


φ -Spirale als optimale
Roboter-Trajektorie

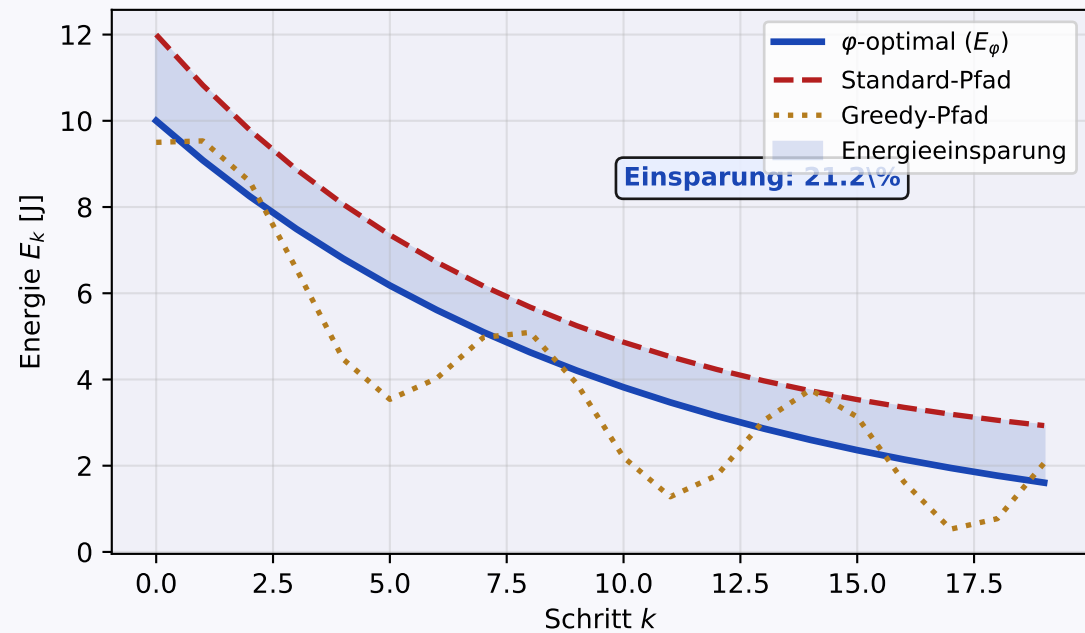


Goldener Schnitt φ in Roboterkinematik und Pfadoptimierung

$$(\theta_k = k \cdot 2\pi/\varphi^2 \bmod 2\pi)$$



Energieverbrauch: φ -Pfad vs. Alternativen



Fibonacci-Pfadlängen konvergieren zu φ

